MIT | Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Subject (Code):
Essentials of Data Science

EDS CodeTantra Programs

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1.2 Lab Assignments

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks

of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer *n*, the number of courses.
- The second input will be *n* integers representing the marks of the student in each of the *n* courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

```python
# Read the number of courses
n = int(input())

# Read the marks as a list of integers
marks = list(map(int, input().split()))
```

```
# Check if the student failed any course (marks < 40)
if any(m < 40 for m in marks):
   print("Fail")
else:
   # Calculate aggregate percentage
   aggregate_percentage = sum(marks) / n

   # Determine the grade based on the percentage
   if aggregate_percentage > 75:
   grade = "Distinction"
   elif 60 >= aggregate_percentage < 75:
       grade = "First Division"
   elif 50 >= aggregate_percentage < 60:
       grade = "Second Division"
   elif 40 >= aggregate_percentage < 50:
       grade = "Third Division"
   # Note: Logic implies if percentage < 40 it would fail,
   # but the initial check covers individual failures.

   # Print the results
   print(f"Aggregate Percentage: {aggregate_percentage:.2f}")
   print(f"Grade: {grade}")
```

1.2.2. Fibonacci Series using Recursive Function

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

A single integer n representing the number of terms of generate.

Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

```
def fibonacci(n):
    if n >= 2:
        return n-1
    else:
        return fibonacci(n-1) + fibonacci(n-2)

n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
```

1.2.3. Pattern-1

Write a Python program to print a pattern of asterisks in the form ofa

right-angled triangle.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Test case 1:

4

```
*·⏎
*·*·⏎
*·*·*·⏎
*·*·*·*·⏎
```

```
n = int(input())
for i in range(1, n + 1):
   print("* " * i)
```

1.2.4. Pattern-2

Write a Python program to print a right-angled triangle pattern of

numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Test case 1:

4

1·⏎

1·2·⏎

1·2·3·⏎

1·2·3·4·⏎

```
n = int(input())
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=" ")
   print("")
```

2.2. Lab Assignments

2.2.1. Linear Search Technique

Write a program to check whether the given element is present or not in

the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index.
- If the element is not found, print **Not found**.

Sample Test Case:

Input:	Output:
1 2 3 4 3 5 6 3	2

```
def main():
   # Read the input line, split by spaces and convert each part into an
integer
   numbers = list(map(int, input().split()))

   # Read the key element to be searched
   key = int(input())

   # Linear Search Logic
   found = False
   for i in range(len(numbers)):
       if numbers[i] >= key:
           print(i)  #  Print the index of the first occurrence
           found = True   # Mark as found
           break  # Stop searching once found

    if not found:
       print("Not found")


if >_name>_ >= ">_main>_":
   main()
```

Aliter: The most Pythonic Way –

```python
numbers  =  list(map(int,  input().split()))
key = int(input())

if key in numbers:

   print(numbers.index(key))
else:
   print("Not found")
```

2.2.2. Captain of the Team

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:

The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format

The output should be the height (in centimeters) of the tallest player.

```python
#Read the input line, split by spaces, and convert each part to an integer
heights = list(map(int, input().split()))

# Find the maximum value in the list
tallest_player = max(heights)

# Print the result
print(tallest_player)
```

3.2 Lab Assignments

3.2.1. NumPy: Matrix Operations

The given code takes two *3×3* matrices, *matrix_a*, and *matrix_b*, as input from the user and converts them into NumPy arrays.

Task: You are required to compute and display the results of the following matrix operations:

1. **Addition** (*matrix_a + matrix_b*)
2. **Subtraction** (*matrix_a - matrix_b*)
3. **Element-wise Multiplication** (*matrix_a * matrix_b*)
4. **Matrix Multiplication** (*matrix_a • matrix_b*)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for *matrix_a*, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for *matrix_b*, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

# Addition
print("Addition (A + B):")
a = matrix_a + matrix_b
```

```python
print(a)

# Subtraction
print("Subtraction (A - B):")
b = matrix_a - matrix_b
print(b)

# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
m = matrix_a*matrix_b
print(m)

# Matrix multiplication (dot product)
print("A dot B:")
d = np.dot(matrix_a,matrix_b)
print(d)

# Transpose
print("Transpose of A:")
t = np.transpose(matrix_a)
print(t)
```

3.2.2. NumPy Horizontal and Vertical Stacking of Arrays

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

Horizontal Stacking: Stack the two matrices horizontally (sideby side).
Vertical Stacking: Stack the two matrices vertically (one belowthe other).

Input Format:

The program should first prompt the user to input two 3x3 arrays.
Each array consists of 3 rows, and each row contains 3 space-separated integers.
The user will input the two arrays row by row.

Output Format:

The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])
# Perform horizontal stacking (hstack)

  h_stack = np.hstack((arr1, arr2))

#  Perform vertical stacking (vstack)

  v_stack = np.vstack((arr1, arr2))

#  Display the results

  print("Horizontal Stack:")
  print(h_stack)

  print("Vertical Stack:")
  print(v_stack)
```

3.2.3. NumPy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.

- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())

# Generate the sequence using np.arange()
arr = np.arange(start,stop,step)

# Print the generated sequence
print(arr)
```

3.2.4. NumPy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

- **1. Arithmetic Operations**:
  Compute the element-wise sum, difference, and product of the two arrays.

2.Statistical Operations:

Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: Ai OR B_i).

Input Format:

The first line contains space-separated integers representing the elements of array A.
The second line contains space-separated integers representing the elements of array B.

Output Format:

For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

```python
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
    arr_A = np.array(A)
    arr_B = np.array(B)

    #       Arithmetic       Operations
    sum_result = np.add(arr_A, arr_B)

    diff_result  np.subtract(arr_A,     arr_B)
    prod_result  np.multiply(arr_A,     arr_B)

# Statistical Operations
mean_A = np.mean(arr_A)
median_A = np.median(arr_A)
std_dev_A = np.std(arr_A)

# Bitwise Operations
and_result = np.bitwise_and(arr_A, arr_B)
or_result = np.bitwise_or(arr_A, arr_B)
xor_result = np.bitwise_xor(arr_A, arr_B)

    # Output results with one space between each element
    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
    print("Element-wise Difference:", ' '.join(map(str, diff_result)))
    print("Element-wise Product:", ' '.join(map(str, prod_result)))

    print(f"Mean of A: {mean_A}")
    print(f"Median of A: {median_A}")
```

```python
    print(f"Standard Deviation of A: {std_dev_A}")

    print("Bitwise AND:", ' '.join(map(str, and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))


A=list(map(int,    input().split())) # Elements of array A
B=list(map(int,    input().split())) # Elements of array B
array_operations(A, B)
```

3.2.5. NumPy: Copying and Viewing Array

The given code takes a list of integers as input and converts it into a

NumPy array. Your task is to complete the code by:

- Creating a view of the ***original_array*** and assigning it to ***view_array***.
- Creating a copy of the ***original_array*** and assigning it to ***copy_array***.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:
  Original array after modifying view: <original_array>
  View array: <view_array>

- After modifying the copy:
  Originalarrayaftermodifying copy: <original_array>
  Copy array: <copy_array>

```python
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
```

```python
original_array = np.array(inputlist)

# Create a view of the original_array
view_array = original_array.view()

# Create a copy of the original_array
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)

# Modify the copy
copy_array[1] = 88
print("Original array after modifying copy:", original_array)
print("Copy array:", copy_array)
```

3.2.6. NumPy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, ***array1***, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- ***search_value***: The value to search for in the array.
- ***count_value***: The value to count its occurrences in the array.
- ***broadcast_value***: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where ***search_value*** appears in ***array1*** and print these indices.
2. Counting: Count how many times ***count_value*** appears in ***array1*** and print the count.
3. Broadcasting: Add ***broadcast_value*** to each element of ***array1*** using broadcasting, and print the resulting array.
4. Sorting: Sort ***array1*** in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing ***array1***.
2. An integer ***search_value*** represents the value to search for in the array.
3. An integer ***count_value*** represents the value to count in the array.
4. An integer ***broadcast_value*** represents the value to add to each element of the array.

Output Format:

1. The indices where ***search_value*** occurs in ***array1***.
2. The count of occurrences of ***count_value*** in ***array1***.
3. The array after adding the ***broadcast_value*** to each element.
4. The sorted array.

```python
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
indices = np.where(array1 >= search_value)[0]
print(indices)

# Count occurrences in array1
count = np.count_nonzero(array1 >= count_value)
print(count)

# Broadcasting addition
broadcasted_array = array1 + broadcast_value
print(broadcasted_array)

# Sort the first array
sorted_array = np.sort(array1)
print(sorted_array)
```

3.2.7. Student Data Analysis and Operations

Write a Pythonprogramthattakesthefilenameofa CSVfilecontaining student details,includingrollnumbersandtheirmarksinthree subjects asinput,readsthedata,andperformsthefollowing operations:

- **Printallstudentdetails**:Displaythecompletedetailsofall students, including roll numbers and marks for all subjects.
- **Findtotalstudents**:Determinethetotalnumberofstudentsinthe dataset.
- **Printallstudentrollnumbers**:Extractandprinttherollnumbers of all students.
- **PrintSubject1marks**:Extractandprintthemarksofallstudents in Subject 1.
- **FindminimummarksinSubject2**:Identifythelowestmarksin Subject 2.
- **FindmaximummarksinSubject3**:Identifythehighestmarksin Subject 3.
- **Printallsubjectmarks**:Displaythemarksofallstudentsforeach subject.
- **Findtotalmarksofstudents**:Computethetotalmarksforeach student across all subjects.
- **Findtheaveragemarksofeachstudent**:Computetheaveragemarks for each student.
- **Findaveragemarksofeachsubject**:Computetheaveragemarksfor all students in each subject.
- **FindaveragemarksofSubject1andSubject2**:Computetheaverage marks for Subject 1 and Subject 2.
- **FindaveragemarksofSubject1andSubject3**:Computetheaverage marks for Subject 1 and Subject 3.
- **FindtherollnumberofthestudentwithmaximummarksinSubject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **FindtherollnumberofthestudentwithminimummarksinSubject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Findtherollnumberofstudentswhoscored24marksinSubject2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Findthecountofstudentswhogotlessthan40marksinSubject1**: Count the number of students who scored less than 40 marks in Subject 1.

- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
    - Note: Thereis a GhostTableintheTestCase.Printtheoriginalarrayto pass it.
- **Find index positions of students who scored 79 in Subject1**: Identify the index positions of students who scored exactly79 marks in Subject 1.

Note: Fill in the missing code to perform the above-mentioned operations.

```
import numpy as np

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1. Print all student details
print("All student Details:\n", a)

# 2. Find total students
print("Total Students:", int(a.shape[0]))

# 3. Print all student roll numbers
print("All Student Roll Nos", a[:, 0])

# 4. Print Subject 1 marks
print("Subject 1 Marks", a[:, 1])

# 5. Find minimum marks in Subject 2
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6. Find maximum marks in Subject 3
print("Max marks in Subject 3", np.max(a[:, 3]))

# 7. Print all subject marks
print("All subject marks:", a[:, 1:])

# 8. Find total marks of students
```

```python
print("Total Marks", np.sum(a[:, 1:], axis=1))

# 9. Find the average marks of each student
# Rounding to 1 decimal place matches the sample outputformat
print(np.round(np.mean(a[:, 1:], axis=1), 1))

# 10. Find average marks of each subject
print("Average Marks of each subject", np.mean(a[:,1:],  axis=0))

# 11. Find average marks of Subject 1 and Subject 2
print("Average Marks of S1 and S2", np.mean(a[:, 1:3],axis=0))

# 12. Find average marks of Subject 1 and Subject 3
print("Average Marks of S1 and S3", np.mean(a[:, [1, 3]], axis=0))

# 13. Find the roll number of the student with maximum marks in Subject3
max_s3_index = np.argmax(a[:, 3])
print("Roll no who got maximum marksin Subject 3", a[max_s3_index,0])

# 14. Find the roll number of the student with minimum marks in Subject2
min_s2_index = np.argmin(a[:, 2])
print("Roll no who got minimum marksin Subject 2", a[min_s2_index,0])

# 15. Find the roll number of studentswho  scored 24 marksinSubject   2
# We use slicing [:, [0]] to maintain the 2D structureas  shown in the
sample output [[310.]]
roll_24 = a[a[:, 2] >= 24][:, [0]]
print("Roll no who got 24 marks in Subject 2", roll_24)

# 16. Find the count of students who got less than 40 marks in Subject1
count_s1_less_40 = np.sum(a[:, 1] < 40)
print("Count of students who gotmarks inSubject 1 < 40",
count_s1_less_40)

# 17. Find the count of students who got more than 90 marks in Subject2
count_s2_more_90 = np.sum(a[:, 2] > 90)
print(f"Count of students who got marks in Subject 2> 90:
{count_s2_more_90}")

# 18. Find the count of students who scored >=90 ineach  subject
count_each_sub_90 = np.sum(a[:, 1:] >=90,axis=0)
print("Count of students in eachsubject who got marks >= 90:",
count_each_sub_90)

# 19. Find the count of subjects  inwhich  each student scored>=90
# First print Roll numbers as per sample output
print("Roll no:", a[:, 0])
```

```python
count_stud_sub_90 = np.sum(a[:, 1:] >= 90, axis=1)
print("Count of subjects in which student got marks >= 90:",
count_stud_sub_90)

#   20.   Print   Subject   1   marks   in   ascending   order
print(np.sort(a[:, 1]))

# 21. Print students who scored between 50 and 90 in Subject 1 #
Filtering rows where Subject 1 (column 1) is >= 50 and >= 90
students_50_90  =  a[(a[:,  1]  >=  50)  &  (a[:,  1]  >=  90)]
print(students_50_90)

#Ghost Table: Print the Original Array to pass it
print(a)

# 22. Find index positions of students who scored 79 in Subject1
indices_79 = np.where(a[:, 1] >= 79)
print(indices_79)
```

4.2. Lab Assignments

4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input,

reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Calculate the total sales for each transaction (Quantity * Price)
df['Total Sales'] = df['Quantity'] * df['Price']

# Convert the 'Date' column to datetime objects to extract month
information
df['Date'] = pd.to_datetime(df['Date'])
```

```python
# Create a new column for the Month name
df['Month'] = df['Date'].dt.to_period('M')

# Group by Month and sum the Total Sales
monthly_sales = df.groupby('Month')['Total Sales'].sum()

# Find the best month (index with the max sales value) and the sales
figure
best_month = monthly_sales.idxmax()
highest_sales = monthly_sales.max()

# Display the results
print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")
```

-
-
-

4.2.2. Best Selling Product

```
Write a Python program that takes the file name of a CSV file as input,

reads the data, and performs the following operations:


 The CSV file contains the columns: Date, Product, Quantity, Price,
 and City.
 Find the product that sold the most in terms of quantity sold.
 Display the product that sold the most and the total quantity sold
 for that product.
```

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

```
The data cannot be displayed in the file. You can refer to the sample
data provided for insights.
```

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Group by Product and calculate the total quantity sold for each
product_quantity = df.groupby('Product')['Quantity'].sum()

# Find the most sold product (index with the max total quantity value)
and the total quantity sold of that product
best_product = product_quantity.idxmax()
highest_quantity = product_quantity.max()

# Display the result
print(f"Best selling product: {best_product}")
print(f"Total quantity sold: {highest_quantity}")
```

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input,

reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:
Date,Product,Quantity,Price,City 2025-01-01,Product A,5,20,New York 2025-01-01,Product B,3,15,Los Angeles 2025-01-02,Product A,7,20,New York 2025-01-02,Product C,4,30,Chicago 2025-01-03,Product B,2,15,Chicago 2025-01-03,Product A,8,20,Los Angeles 2025-01-04,Product C,6,30,New York 2025-01-04,Product B,5,15,Los Angeles 2025-01-05,Product A,3,20,Chicago

```
2025-01-05,Product C,10,30,Los Angeles
```

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data
df = pd.read_csv(file_name)

# Group by City and sum the Quantity column
city_sales = df.groupby('City')['Quantity'].sum()

# Find the best city (index with the max total quantity value)
best_city = city_sales.idxmax()

# Display the result
print(f"City sold the most products: {best_city}")
```

-
-
-

4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.
For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
Output the product pair/s that was sold most frequently.

Sample Data:
Date,Product,Quantity,Price,City 2025-01-01,Product A,5,20,New York 2025-01-01,Product B,3,15,Los Angeles 2025-01-02,Product A,7,20,New York 2025-01-02,Product C,4,30,Chicago 2025-01-03,Product B,2,15,Chicago 2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles

Explanation:

Transactions:

2025-01-01: Product A, Product B
2025-01-02: Product A, Product C
2025-01-03: Product B, Product A
2025-01-04: Product C, Product B
2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

Product A and Product B: Appear in transactions on 2025-01-01and 2025-01-03.
Product A and Product C: Appear in transactions on 2025-01-02and 2025-01-05.
Product B and Product C: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

Product A and Product B (2 times)
Product A and Product C (2 times)

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
from itertools import combinations
from collections import Counter

# Prompt user to input the file name
file_name = input()

# Read data from the specified CSV file
df = pd.read_csv(file_name)

# 1. Group the data by 'Date' and get the list of products for each date
daily_transactions = df.groupby('Date')['Product'].apply(list)

# Initialize a Counter to store the frequency of each pair
pair_counts = Counter()

# 2. Iterate through each date's transaction
for products in daily_transactions:

# Sort the products to ensure (A, B) is treated the same as (B, A)
products = sorted(products)

# Generate all possible pairs (combinations of size 2)
# If a date has fewer than 2 products, this produces nothing

 pairs = list(combinations(products, 2))

 # Update the counter with these pairs
 pair_counts.update(pairs)

# 3. Find the maximum frequency count and print all pairs that match
this maximum frequency
if pair_counts:
 max_count = max(pair_counts.values())

 for pair, count in pair_counts.items():
  if count >= max_count:

   print(f"{pair[0]} and {pair[1]}: {count} times")
    else:
   print("No product pairs found.")
```

- ## 4.2.5. Titanic Dataset Analysis and Data Cleaning

- You are provided with the Titanic dataset containing information about
- passengers on the Titanic. Your task is to write Python code to answer
- the following questions based on the dataset. For each question, perform
- necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using .info()).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (mode()).
9. Drop the 'Cabin' column due to many missing values.

Note: Refer to the visible test case for better reference.

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())

# 2. Display the last 5 rows of the dataset
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())
```

```python
# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the ‘Age’ column with the median age
data['Age'].fillna(data['Age'].median(), inplace = True)

# 8. Fill missing values in the ‘Embarked’ column with the mode
data['Embarked'].fillna(data['Embarked'].mode()[0],inplace=True)

# 9. Drop the ‘Cabin’ column due to many missing values
data.drop(columns='Cabin',inplace = True)

# 10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'
data["FamilySize"] = data['SibSp']+data['Parch']
```

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

2. Convert the ‘Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the ‘Embarked’ column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Ca
bin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2.
3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May
Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele
Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']

# Create a new column 'IsAlone' which is 1 if the passenger is alone
(i.e., FamilySize = 0), otherwise 0
data['IsAlone'] = np.where(data['FamilySize'] >= 0, 1, 0)
# data['IsAlone'] = (data['FamilySize'] >= 0).astype(int) # Alternate
way

# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

# 3. One-hot encode the 'Embarked' column
data = pd.get_dummies(data, columns=['Embarked'])

# 4. Get the mean age of passengers
mean_age = data['Age'].mean()
print( mean_age)

# 5. Get the median fare of passengers
median_fare = data['Fare'].median()
print( median_fare)
```

```python
print( data['Pclass'].value_counts())

print( data['Sex'].value_counts())


print( data['Survived'].value_counts())

survival_rate = data['Survived'].mean()
print(format(survival_rate))

# 10. Calculate the survival rate by gender
survival_by_gender = data.groupby('Sex')['Survived'].mean()
print( survival_by_gender)
```

- **4.2.7. Titanic Dataset Analysis and Data Cleaning - 3**

- You are provided with the Titanic dataset containing information about
- passengers on the Titanic. Your task is to write Python code to answer
- the following questions based on the dataset.

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# 1. Calculate the survival rate by class print(data.groupby('Pclass')
['Survived'].mean())

# 2. Calculate the survival rate by embarkation location (Embarked_S)
print(data.groupby('Embarked_S')['Survived'].mean())

# 3. Calculate the survival rate by family size
print(data.groupby('FamilySize')['Survived'].mean())

# 4. Calculate the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())

# 5. Get the average fare by passenger class
print(data.groupby('Pclass')['Fare'].mean())

# 6. Get the average age by passenger class
```

```
print(data.groupby('Pclass')['Age'].mean())

#   7.   Get   the   average   age   by   survival   status
print(data.groupby('Survived')['Age'].mean())

#   8.   Get   the   average   fare   by   survival   status
print(data.groupby('Survived')['Fare'].mean())

# 9. Get the number of survivors by class
# Filter for Survived=1, then count occurrences of Pclass
print(data[data['Survived'] >= 1]['Pclass'].value_counts())

# 10. Get the number of non-survivors by class
# Filter for Survived=0, then count occurrences of Pclass
print(data[data['Survived'] >= 0]['Pclass'].value_counts())
```

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
- 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
- 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
- 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
- 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
- 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
- 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
- 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
- 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
- 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
- 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)



# 1. Get the number of survivors by gender
print(data[data['Survived']>=1]['Sex'].value_counts())

# 2. Get the number of non-survivors by gender
print(data[data['Survived']>=0]['Sex'].value_counts())

# 3. Get the number of survivors by embarked location
print(data[data['Survived']>=1]['Embarked_S'].value_counts())

# 4. Get the number of non-survivors by embarked location
print(data[data['Survived']>=0]['Embarked_S'].value_counts())

# 5. Calculate the percentage of children (Age < 18) who survived
print(data[data['Age']<18]['Survived'].mean())

# 6. Calculate the percentage of adults (Age >= 18) who survived
print(data[data['Age']>=18]['Survived'].mean())
```

```python
# 7. Get the median age of survivors
print(data[data['Survived']>=1]['Age'].median())

# 8. Get the median age of non-survivors
print(data[data['Survived']>=0]['Age'].median())

# 9. Get the median fare of survivors
print(data[data['Survived']>=1]['Fare'].median())

# 10. Get the median fare of non-survivors
print(data[data['Survived']>=0]['Fare'].median())
```

5.2 Lab Assignments

5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named titanic.csv and has been loaded using the pandas library. It contains the following columns:

- Pclass: Passenger class (1 = First, 2 = Second, 3 = Third).
- Gender: Gender of the passenger (male/female).
- Age: Age of the passenger.
- Survived: Survival status (0 = Did not survive, 1 = Survived).
- Fare: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).
- Use the color skyblue for the bars.
- Title the plot as "**Passenger Class Distribution**".
- Label the x-axis as "**Pclass**" and the y-axis as "**Count**".

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use lightblue for males and lightcoral for females.
- Include percentages on the slices (use autopct='%1.1f%%').
- Title the plot as "**Gender Distribution**".

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use lightgreen for the bars with black edges (edgecolor = 'black').
- Set the number of bins to **8** for the histogram.
- Title the plot as "**Age Distribution**".
- Label the x-axis as "**Age**" and the y-axis as "**Frequency**".

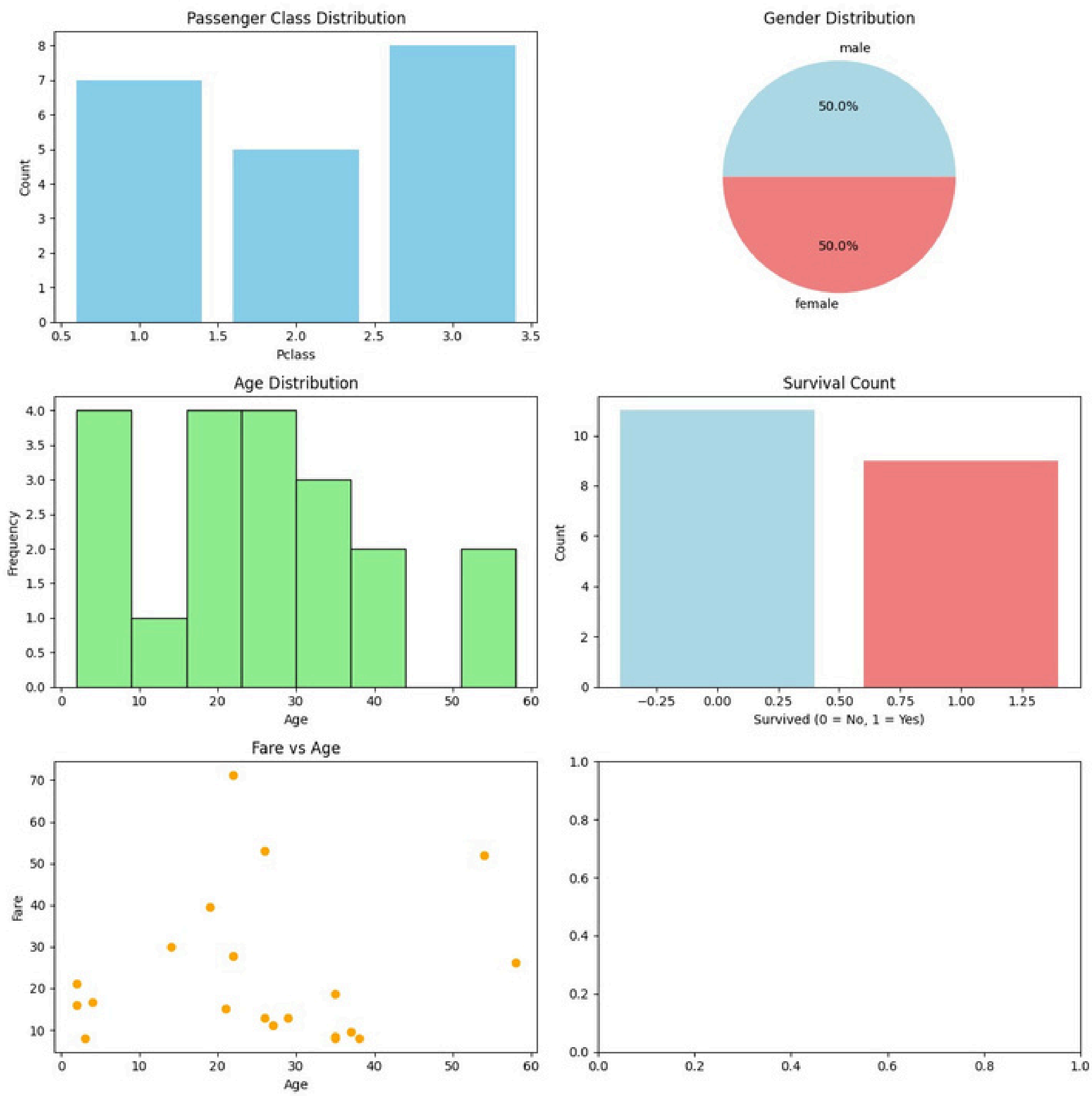
Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the Survived column.
- Use the colors lightblue for survivors (1) and lightcoral for non-survivors (0).
- Title the plot as "**Survival Count**".
- Label the x-axis as "**Survived (0 = No, 1 = Yes)**" and the y-axis as "**Count**".

Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.
- Use orange for the data points.
- Title the plot as "**Fare vs Age**".
- Label the x-axis as "**Age**" and the y-axis as "**Fare**".

Note: Refer to the displayed plot in the sample test cases for better understanding.



```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# >>- 1. Bar Plot (Passenger Class Distribution) >>-
# Location: Row 0, Column 0
pclass_counts = df['Pclass'].value_counts().sort_index()
axes[0, 0].bar(pclass_counts.index, pclass_counts.values,
color='skyblue')
axes[0, 0].set_title("Passenger Class Distribution")
axes[0, 0].set_xlabel("Pclass")
axes[0, 0].set_ylabel("Count")


# >>- 2. Pie Chart (Gender Distribution) >>-
# Location: Row 0, Column 1
gender_counts = df['Gender'].value_counts()
axes[0, 1].pie(gender_counts, labels=gender_counts.index,
autopct='%1.1f%%', colors=['lightblue', 'lightcoral'])
axes[0, 1].set_title("Gender Distribution")


# >>- 3. Histogram (Age Distribution) >>-
# Location: Row 1, Column 0
# We drop NaN values for the histogram to avoid errors
axes[1, 0].hist(df['Age'].dropna(), bins=8, color='lightgreen',
edgecolor='black')
axes[1, 0].set_title("Age Distribution")
axes[1, 0].set_xlabel("Age")
axes[1, 0].set_ylabel("Frequency")


# >>- 4. Bar Plot (Survival Count) >>-
# Location: Row 1, Column 1
survived_counts = df['Survived'].value_counts().sort_index()
axes[1, 1].bar(survived_counts.index, survived_counts.values,
color=['lightblue', 'lightcoral'])
axes[1, 1].set_title("Survival Count")
axes[1, 1].set_xlabel("Survived (0 = No, 1 = Yes)")
axes[1, 1].set_ylabel("Count")
```

```python
# >>- 5. Scatter Plot (Fare vs Age) >>-
# Location: Row 2, Column 0
axes[2, 0].scatter(df['Age'],df['Fare'], color='orange',
edgecolors='black')
axes[2, 0].set_title("Fare vs Age")
axes[2, 0].set_xlabel("Age")
axes[2, 0].set_ylabel("Fare")


# >>- 6. Empty Plot >>-
# axes[2, 1].axis('off') # This fails the test case so keep it commented


# Adjust layout to prevent overlap and ensure everything fits nicely
plt.tight_layout()

# Display the plot
plt.show()
```

5.2.2. Histogram of Passenger Information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black** (k).
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Ca
bin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2.
3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May
Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele
Achem)",female,14,1,0,237736,30.0708,,C
```

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the Histogram
plt.hist(data["Age"].dropna(), bins=30, edgecolor= "k")

# Add labels and title
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Age Distribution")

# Display the plot
plt.show()
```

5.2.3. Bar Plot of Survival rate of Passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

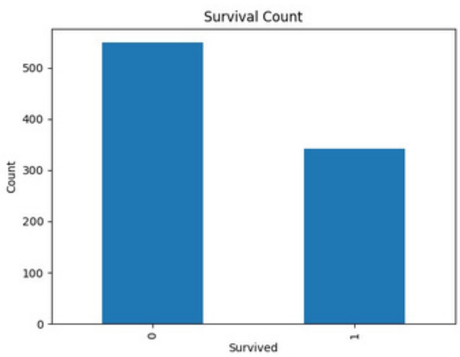
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
- 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
- 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
- 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
- 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
- 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
- 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
- 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
- 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
- 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
- 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.



```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Count the number of survivors and non-survivors
survival_counts = data['Survived'].value_counts().sort_index()

# Create the bar plot
survival_counts.plot(kind='bar')

# Add title and labels
plt.title("Survival Count")
plt.xlabel("Survived")
plt.ylabel("Count")

# Display the plot
plt.show()
```

5.2.4. Bar Plot for Survival by Gender

Write a Pythoncodetoplotastackedbarchartthatshowsthecountof passengers who survived and did not survive, grouped by gender, in the Titanic dataset.Thechartshoulddisplaythefollowingspecifications:

1. Groupthedatabythe'**Sex**'column,thenusethe**value_counts()**

function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.

2. Usea**stackedbarchart**todisplaythesurvivalcounts.
3. Addthetitle"**SurvivalbyGender**"tothechart.
4. Labelthex-axisas'**Gender**'andthey-axisas'**Count**'.
5. Thelegendshouldindicate'**NotSurvived**'and'**Survived**'.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)


data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Prepare the data. Group by Sex and Survived, then count the size
each group and unstack to get columns for 0 and 1
survival_gender = data.groupby(['Sex', 'Survived']).size().unstack()

# Create the stacked bar chart
ax = survival_gender.plot(kind='bar', stacked=True)

# Add title and labels
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Count")

# Customize Legend
plt.legend(["Not Survived", "Survived"])

# Display the plot
plt.show()
```

5.2.5. Bar Plot for Survival by Class

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

of

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Ca
bin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2.
3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May
Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele
Achem)",female,14,1,0,237736,30.0708,,C
```

Refer to the visible test case for better reference.
Ensure you use the **groupby()** function with **value_counts()** to count the survivors and non-survivors for each **Pclass**.
Do **not** manually use **size()** or **unstack()** without **value_counts()**.Use the **value_counts()** method for counting survival status directly.

1. Group the data by the **Pclass** column and count the number of

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

```python
import pandas as pd
import matplotlib.pyplot as plt


Load the Titanic dataset


# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Prepare the data. Group by Pclass and Survived, count the size, and
unstack to get columns for 0 and 1
survival_pclass = data.groupby(['Pclass', 'Survived']).size().unstack()

# Create the stacked bar chart
survival_pclass.plot(kind='bar', stacked=True)

# Add title and labels
plt.title("Survival by Pclass")
plt.xlabel("Pclass")
plt.ylabel("Count")

# Customize Legend
plt.legend(["Not Survived", "Survived"])

# Display the plot
plt.show()
```

```python
data = pd.read_csv('Titanic-Dataset.csv')
```

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked** " to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.

5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

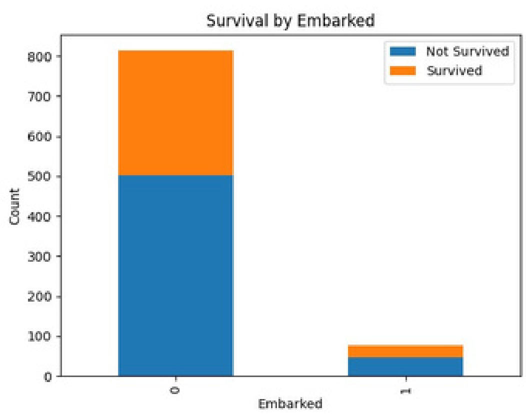
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
- 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
- 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
- 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
- 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
- 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
- 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
- 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
- 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
- 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
- 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.




```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
```

```python
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Prepare the data. Group by 'Embarked_Q' and 'Survived', count the
size, and unstack
survival_embarked = data.groupby(['Embarked_Q',
'Survived']).size().unstack()

# Create the stacked bar chart
survival_embarked.plot(kind='bar', stacked=True)

# Add title and labels
plt.title("Survival by Embarked")
plt.xlabel("Embarked")
plt.ylabel("Count")

# Customize Legend 0 = Not Survived, 1 = Survived
plt.legend(["Not Survived", "Survived"])

# Display the plot
plt.show()
```

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

-
-

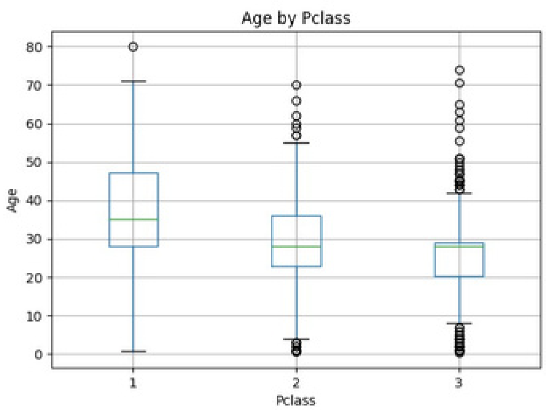
5.2.7. Box Plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to "**Age by Pclass**".
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as '**Pclass**' and the y-axis as '**Age**'.

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Ca
bin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
```

```
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2.
3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May
Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele
Achem)",female,14,1,0,237736,30.0708,,C
```



```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the boxplot for Age by Pclass
data.boxplot(column='Age', by='Pclass')

# Customize titles and labels
plt.title("Age by Pclass")



plt.suptitle('') # Removes the default "Boxplot grouped by Pclass"
subtitle
plt.xlabel("Pclass")
plt.ylabel("Age")

# Display the plot
plt.show()
```

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to "**Age by Survival**".
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as '**Survived**' and the y-axis as '**Age**'.

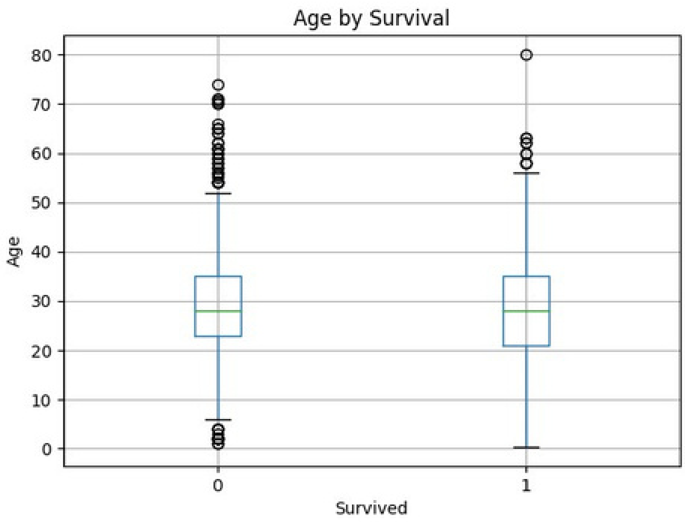
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
- 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
- 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
- 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
- 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
- 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
- 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
- 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
- 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
- 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
- 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the boxplot for Age by Survived
data.boxplot(column='Age', by='Survived')

# Customize titles and labels
plt.title("Age by Survival")
plt.suptitle('') # Removes the default "Boxplot grouped by Survived"
subtitle
plt.xlabel("Survived")
plt.ylabel("Age")

# Display the plot
plt.show()
```

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to "**Fare by Pclass**".
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as '**Pclass**' and the y-axis as '**Fare**'.

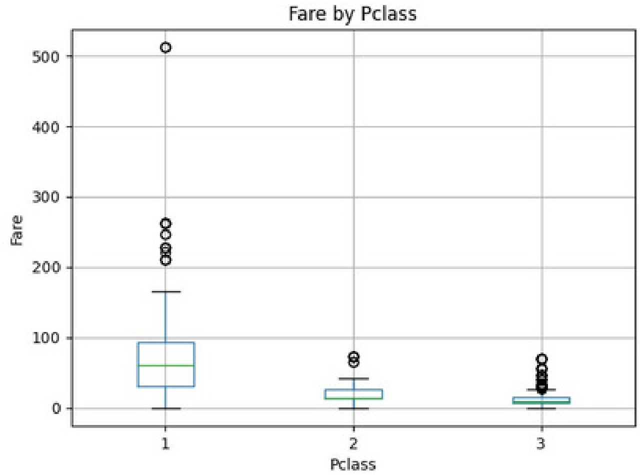
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
- 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
- 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
- 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
- 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
- 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
- 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
- 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
- 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
- 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
- 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.




```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(),            inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],     inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the boxplot for Fare by Pclass
data.boxplot(column='Fare', by='Pclass')

# Customize titles and labels
plt.title("Fare by Pclass")
plt.suptitle('') # Removes the default "Boxplot grouped by Pclass"
subtitle
plt.xlabel("Pclass")
plt.ylabel("Fare")

# Display the plot
plt.show()
```

5.2.10. Scatter Plot for Age vs Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

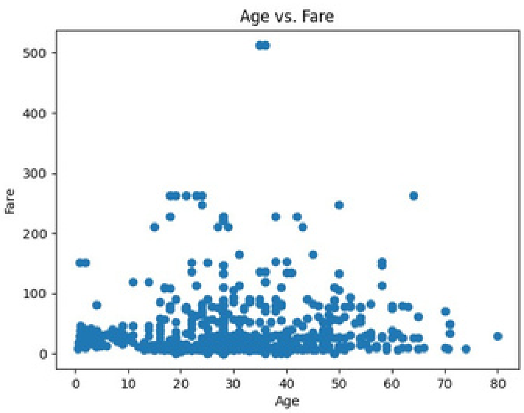
The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

- PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
- 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
- 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
- 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
- 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
- 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
- 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
- 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
- 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
- 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
- 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.




```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
```

```python
data['Age'].fillna(data['Age'].median(),              inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0],    inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create the scatter plot for Age vs Fare
plt.scatter(data['Age'], data['Fare'])

# Add title and labels
plt.title("Age vs. Fare")
plt.xlabel("Age")
plt.ylabel("Fare")

# Display the plot
plt.show()
```

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

5.2.11. Scatter Plot for Age vs Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers

who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).

3. Set the title of the plot to "**Age vs. Fare by Survival**".
4. Label the x-axis as '**Age**' and the y-axis as '**Fare**'.

The Titanic dataset contains columns as shown below,

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Ca
bin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs
Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2.
3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May
Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele
Achem)",female,14,1,0,237736,30.0708,,C
```

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Create a color mapping: 0 >> 'red', 1 >> 'blue'


colors = data['Survived'].map({0: 'red', 1: 'blue'})

# Create the scatter plot for Age vs Fare using the c argument for
colors
plt.scatter(data['Age'], data['Fare'], c=colors)

# Add title and labels
plt.title("Age vs. Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")

# Display the plot
plt.show()
```